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MATHEMATICS
ESSAY PROBLEMS

Name : _____ **Index Number** : _____

Country : _____



18th International Mathematics and Science Olympiad

Indonesia

15 December 2021

Instructions:

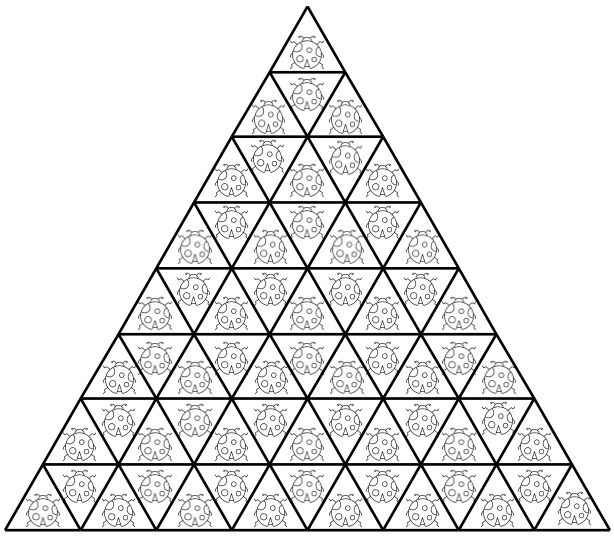
1. Write your name, country and index number on every page of this booklet.
2. Write your detailed solution in Arabic Numerical or English in the space provided for each question in this booklet. If you need more space for your working solutions, you may use the reverse side of each page, but please write "next side continued" in the last line.
3. There are 13 questions in this paper.
4. Each question is worth 3 marks and partial credits may be awarded.
5. Diagrams are NOT drawn to scale. They are intended only as aids.
6. You have 90 minutes to complete this paper.
7. Use black pen or blue pen or pencil to write your answer.

Do not turn over this page until you are told to do so.

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

1. In the diagram below, an equilateral triangular carpet of side length 8 is divided into 64 triangular cells, where in each cell, there is one ladybug. At a go signal, each ladybug moves to a neighboring cell, which is a cell having a common side with its current cell. After all the ladybugs have moved, what is the minimum number of empty cells that can be found in the carpet? *Note*: After the move, any cell can have more than one ladybug occupying it.



Answer : _____ empty cells

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

2. There are ten switches as shown below.



A B C D E F G H I J

It is known that five of them are working, while the other five are not. The light will only turn on if the number of working switches that are turned on is greater than the number of those that are not. Seven children tried to turn on the lights, and each child tried it when all the switches were turned off.

Ali = Turned on switch A, B, C
Budi = Turned on switch B, C, D
Charly = Turned on switch C, D, E
Doni = Turned on switch A, E, H, I, J
Eren = Turned on switch C, D, E, F, H
Farah = Turned on switch A, D, G, H, I
Galih = Turned on switch A, B, G, H, J

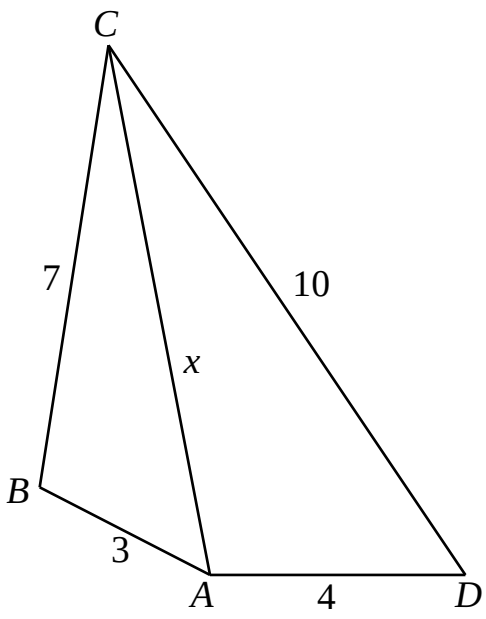
As a result, every child can turn on the lights. If switch J is known to be a working switch, what are the other four working switches? Justify your answer.

Answer : _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

3. Find all possible integers x such that there exist a quadrilateral $ABCD$ with $AB = 3\,\text{cm}$, $BC = 7\,\text{cm}$, $CD = 10\,\text{cm}$, $DA = 4\,\text{cm}$ and $AC = x\,\text{cm}$.



Answer: $x =$ _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

4. Tom tries to construct numbers by substituting letters with digits in the word "*ILOVEIMSO*" in such a way that the following conditions are met:

- The resulting number is divisible by 15.
- Tom uses the same digits for same letters and different digits for different letters.
- Tom does not substitute the letter "*I*" by the digit "0".

What is the difference between the largest possible number and the smallest possible number that can be formed?

Answer： _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

5. A positive integer is called an "*IMSO*" number if it satisfies the following three properties:

- It is a product of two different 2-digit numbers.
- It has exactly 4 factors.
- The difference between the largest and the smallest factors is divisible by the positive difference of other two factors.

For example, 2021 is an *IMSO* number since $2021 = 43 \times 47$ is a product of two different 2-digit numbers, exactly has four factors, namely 1, 43, 47 and 2021, and $2021 - 1 = 2020$ is divisible by $47 - 43 = 4$.

What is the largest 3-digit *IMSO* number?

Answer：_____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

6. Students and teachers from three different grade levels of a school take buses in an excursion. The number of passengers, including teachers, of each grade level are 78, 96, and 85, respectively. There are enough buses of three types: small, medium or large. The information about the different buses is shown in the table below:

Type	Capacity	Fare of each bus
Large bus	15 passengers	$ 12
Medium bus	12 passengers	$ 10
Small bus	10 passengers	$ 9

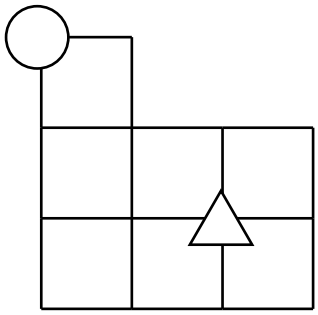
If each grade level should use only one type of bus and any two grades cannot use the same type of bus and each passenger is required to have a seat, what is the total minimum fare?

Answer： $ _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

7. A circle and a triangle are placed at intersections of the grid, where the circle and the triangle are not allowed to lie on the same vertical and horizontal line.
How many different possible ways are there of placing the circle and the triangle?
The diagram shown below is an example.



Answer : _____ ways

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

8. Joseph has received a brand new 2022 calendar planner. The month, date and day of the week are printed on the calendar planner. He was reluctant to use it for a while, and wanted to keep it for a later year. When will be the earliest year that the month, date and day of the week match those on this planner?

Answer : _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

9. Let a, b and c be three positive digits and x be a positive integer. Find the sum of all possible values of $\overline{abc}$, such that $\left(a \times b \times c \times \frac{x}{x+1}\right)^2 = \overline{abc}$.

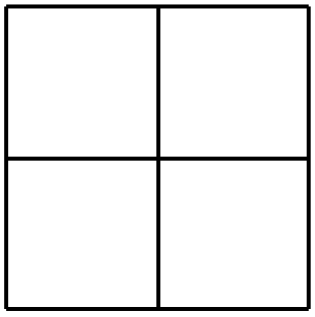
(Note: a, b and c are not necessarily distinct)

Answer：_____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

10. A 2×2 square garden as shown in the diagram below, where humans and animals can only move along the gridlines. A zookeeper is trying to recapture an escaped monkey. Both must stay on the paths. The zookeeper's maximum speed is equal to 2.5 times that of the monkey. They can only always see each other, and can react instantaneously to the action of the other. If the monkey and the zookeeper are in the same segment, the zookeeper can shoot the monkey with a stun gun. Show that the monkey will always be shot. Justify your answer.



ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

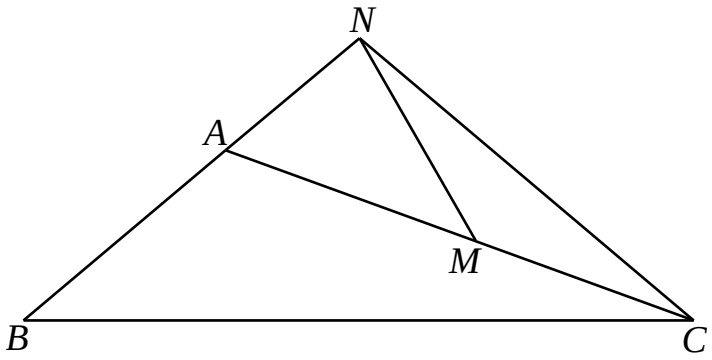
11. When Nala calculated the value of $1^3+2^3+3^3+...+19^3+20^3$, he made a mistake by omitting three cubic numbers. It is known that the correct answer is 44100, while Nala's answer is 40860. If the three cubic numbers are a^3, b^3 and c^3, find $a\times b\times c$.

Answer : _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

12. In the diagram below, ABC is a triangle with $\angle BAC = 120°$ and $\angle ABC = 40°$. M is a point on AC such that $AM = AB$ and N is a point on the extension of BA such that $\angle AMN = 40°$. What is the angle measure, in degrees, of $\angle BNC$?



°

Answer： _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

13. The integers 1, 2, ..., 12 are written on individual cards which are divided among Alice, Boris, Cindy and David, where each person sees only the three cards he or she gets. David is given the correct information that one of Alice's numbers is 8, Boris' numbers are among {2, 3, 5, 7, 11} and Cindy's numbers are either all among {2, 4, 6, 10, 12} or all among {3, 6, 9, 12}. David is then able to determine everyone's numbers. What are Alice's three numbers? Justify your answer.

Answer：_____